

AYE
TECH HUB

• AI & PRODUCTIVITY • LESSON 04

Advanced Prompt Engineering

Chains · Structured Output · Agents · Long Docs

Lesson 04 of 30 | AI & Productivity Program

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- Prompt chaining — break complex tasks into linked steps
- Structured output: get JSON, tables and lists every time
- Working with long documents inside AI context windows
- Meta-prompting and self-critique loops for better results
- Building your first custom AI agent with a system prompt

01 | Prompt Chaining — Break Big Tasks Into Steps

The biggest mistake advanced AI users make is writing one enormous, complex prompt and expecting a perfect result. The better approach is **prompt chaining** — breaking the task into a sequence of smaller prompts where the output of each step becomes the input for the next. This produces dramatically better results for research, writing, analysis, and multi-step reasoning tasks.



Figure 1 — Prompt Chaining: Output of Each Step Feeds the Next

Chain Type	How It Works	Best For
Sequential chain	Output A → input B → output B → input C Each step builds on the previous	Long-form writing, research reports, code generation with testing
Branching chain	One prompt generates multiple outputs that are processed separately then combined	Analysing pros and cons, generating multiple options, parallel research
Iterative refinement chain	Same prompt repeated with instructions to improve the previous output at each cycle	Polishing writing, improving code quality, refining analysis
Validation chain	Generate content → ask AI to fact-check or critique its own output → revise accordingly	Technical accuracy, consistency checks, argument quality

Real Engineering Example — Chained Report Writing

Prompt 1: "Summarise the key challenges of HVAC load calculation in 5 bullets." → Prompt 2: "Using those bullets as headings, write an outline for a 4-section technical report." → Prompt 3: "Write Section 1 using the outline. Formal tone, 300 words." → Prompt 4: "Review Section 1 for technical accuracy and rewrite any weak parts."

02 | Structured Output — Get JSON, Tables and Lists

When you need AI output that your code, spreadsheet, or workflow can process automatically, you need **structured output**. By explicitly requesting a format in your prompt, you can make AI return JSON objects, markdown tables, numbered lists, CSV data, or any other parseable structure — every single time.

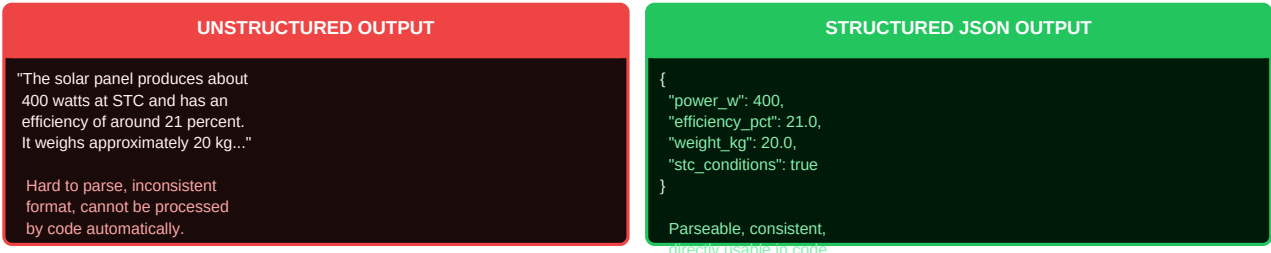


Figure 2 — Unstructured vs Structured (JSON) AI Output

Format	Prompt Instruction	Use Case
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JSON object	"Return your answer as a JSON object with keys: [list keys]. No other text."	Database entry, API integration, automated data processing
JSON array	"Return a JSON array of objects, each with fields: [fields]. No markdown."	Product lists, equipment schedules, comparison datasets
Markdown table	"Format your response as a markdown table with columns: [columns]."	Comparison tables, specifications, schedules for docs
Numbered list	"Provide exactly N items, each on a new line starting with the number."	Steps, recommendations, requirements, options
CSV data	"Output as CSV with headers on the first row. No explanation text."	Data import to Excel, databases, analysis tools
Structured report	"Use exactly these section headers: [headers]. One paragraph per section."	Consistent report formats, documentation templates

03 | Working With Long Documents

Every AI model has a **context window** — the maximum amount of text it can process in one conversation. GPT-4o: ~128K tokens. Claude 3.5: ~200K tokens. Gemini 1.5 Pro: ~1M tokens. (1 token ≈ 0.75 words.) When your document exceeds the context window, or when you want the best results, use these strategies.

Strategy	How to Apply	Best For
Chunking	Split the document into sections. Process each chunk with the same prompt. Combine outputs in a final "synthesis" prompt.	Books, long reports, legal documents, technical manuals
Summarise then query	First prompt: "Summarise this document in 500 words, preserving all key facts." Second prompt: Use the summary as context for detailed questions.	Research papers, meeting transcripts, contracts
Extract then analyse	First prompt: "Extract all [data type] from this document as a list." Second prompt: "Analyse this list and identify patterns/recommendations."	Financial reports, technical specs, survey responses
Rolling window	Process page 1-5 → summarise → process pages 3-8 (overlap) → continue. Keep a running summary that carries context forward.	Narrative documents, policy documents, code repositories
Retrieve and embed	Use a RAG (Retrieval-Augmented Generation) tool like NotebookLM, Claude Projects, or a vector database to query specific sections.	Large knowledge bases, ongoing research, company documentation

04 | Meta-Prompting and Self-Critique Loops

The most powerful advanced technique is asking the AI to **critique and improve its own output**. Research shows that AI models consistently produce better results when prompted to review and revise their first attempt. This mirrors the human writing process: first draft, then edit.

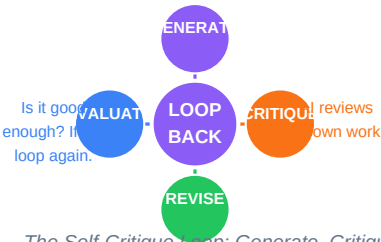


Figure 3 — The Self-Critique Loop: Generate, Critique, Revise

Technique	Prompt Template	Effect
Self-critique	"Review your previous response. Identify 3 weaknesses or gaps. Then rewrite it addressing those weaknesses."	Typically improves depth, accuracy and completeness in one step
Steelmanning	"You just argued [position]. Now give the strongest possible counterargument to your own answer."	Exposes weaknesses and blind spots in AI-generated arguments
Expert review	"Review this as a [field] expert. Would a professional be satisfied? What is missing or incorrect?"	Adds domain-specific depth missing from general responses

Adversarial check	"Try to find a way this plan could fail. What are the 5 most likely failure modes?"	Improves risk analysis, engineering designs, business plans
Meta-prompt	"Before answering, write the ideal prompt for this question. Then answer using that improved prompt."	Forces the model to clarify the problem before solving it

The Two-Step Critique Template

Step 1: "Answer the following question: [your question]" Step 2: "Now review your answer. Rate it 1-10 for accuracy, completeness, and clarity. List 3 specific improvements. Then rewrite it with those improvements applied." This alone improves most AI responses by 30-50%.

05 | Building a Custom AI Agent with a System Prompt

A **custom AI agent** is simply an AI model given a persistent system prompt that defines its role, rules, knowledge, and behaviour. You do not need to write code — you write a thorough system prompt. This is the foundation of custom GPTs, Claude Projects, and every AI assistant built on top of a foundation model.

System Prompt Element	What to Write	Example
Identity and role	Who the agent is, what it specialises in, and its name if it has one	"You are SolarCalc, an expert solar PV system design assistant. You specialise in off-grid and hybrid systems for African climates."
Knowledge scope	What topics it knows about and what it should defer on	"You have deep knowledge of IEC 62109, IEC 61730, and SANS 10142. For medical or legal topics, refer the user to a qualified professional."
Behaviour rules	How it should communicate — tone, format, length defaults	"Always show formulas before calculations. Use metric units. Format answers with headers for questions longer than 3 parts."
Constraints and limits	What it must never do, what assumptions it should always make	"Never recommend specific products by brand name. Always ask for location and load requirements before sizing."
Escalation path	When to tell the user it cannot help and where to send them	"If the user asks for a full system design, recommend they hire a certified PV installer. Direct them to SAPVIA.co.za for Africa."

06 | Advanced Prompting Quick-Reference

Technique	One-Line Template	When to Use
Prompt chaining	Run 3-5 sequential prompts where each output feeds the next	Any task too complex for one prompt: reports, research, code
Structured JSON output	"Return as JSON with keys: [keys]. No other text."	When output needs to be parsed by code or pasted into a tool
Self-critique loop	"Review your answer, rate it, list 3 improvements, rewrite it."	Any important deliverable — always worth the extra step
Role + audience combo	"You are [expert]. Explain [topic] to [audience] in [format]."	Teaching, documentation, audience-specific communication
Constraint forcing	"Answer in exactly [N] words/sentences/bullets. No more, no less."	Focused responses, slide content, social media posts
Socratic mode	"Do not answer directly. Ask me 3 clarifying questions first."	Complex decisions where the AI needs more context from you
Persona + debate	"Argue both sides of [topic] as two experts disagreeing."	Understanding multiple perspectives on complex issues
Document Q&A;	"Here is a document: [paste]. Answer only from this document."	Grounding AI in specific source material, avoiding hallucination
Step-by-step forcing	"Solve this step by step. Show every step before the answer."	Maths, logic puzzles, debugging, multi-step engineering calcs

Format mirroring	"Here is an example of the format I want: [example]. Now do: [task]"	Any time consistency of output format is critical
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What comes next — AI-005

Claude AI for Productivity: a deep dive into Claude's unique strengths (long documents, coding, analysis, safety), Claude Projects, and how to build a personal productivity system using Claude.